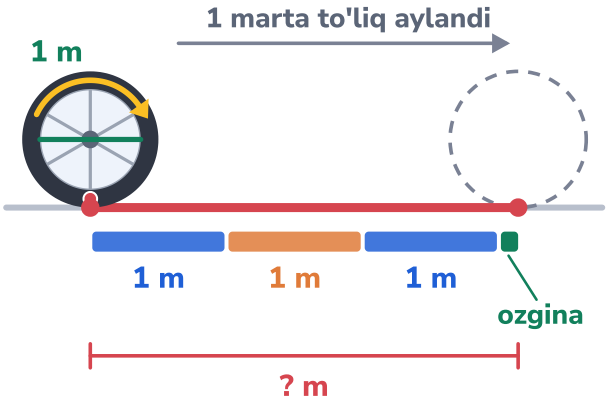
Ratsional, irratsional va haqiqiy sonlar

1-QISM

Qiziqtiramiz

IKKI SAVOL



1-savol. Aravacha g'ildiragining diametri 1 m. G'ildirak bir marta to'liq aylanganda aravacha qancha yo'l bosadi? Rasmda ko'rinib turibdi: yo'l uchta diametrdan sal uzun. Bu yo'lni 3,14 m deymizmi yoki $\frac{22}{7}$ m? Uni oddiy kasr bilan aniq yozib bo'ladimi?

2-savol. Shoira 2 m lentani 3 ta teng bo'lakka qirqdi. Kalkulyatorda $2 : 3$ ni hisoblasa, ekranda 0,6666666667 chiqdi. Bir bo'lak aynan 0,6666666667 m mi?



Kalkulyator ekraniga 10 ta raqam sig'adi. Ekran uzunroq bo'lsa, raqamlar tugarmidi?

ESLAB OLAYLIK

- Kasr — bo'lish: $\frac{3}{4} = 3 : 4 = 0{,}75$. O'nli kasr — oddiy kasr: $0{,}6 = \frac{6}{10} = \frac{3}{5}$.
- Butun sonlar: ..., −3, −2, −1, 0, 1, 2, 3, ... Natural sonlar: 1, 2, 3, ...
- $\sqrt{81} = 9$, chunki $9 \cdot 9 = 81$. $\sqrt{50}$ esa natural son emas: $7 \cdot 7 = 49$, $8 \cdot 8 = 64$.

2-QISM

Sinab ko'ramiz

1-TOPSHIRIQ · KASR KO'RINISHIDA

Har bir sonni ikki butun sonning bo'linmasi — kasr ko'rinishida yozing (maxraj 0 bo'lmasin):

a) 9 b) −4 c) 0 d) $1\frac{3}{5}$ e) 0,7 f) −2,25

Shunday yozib bo'lmaydigan son chiqdimi?

2-TOPSHIRIQ · QOLDIQLARNI KUZATING

Burchak usulida bo'ling. Har qadamdagi qoldiqni yozib boring.

a) $3 : 16$ b) $2 : 11$ c) $5 : 6$

Qaysi birida bo'lish tugaydi? Tugamaganlarida qaysi qoldiq yana chiqdi? Shundan keyin bo'linmada nima bo'ladi?

3-TOPSHIRIQ · KVADRATI 2 GA TENG SON

a) $1{,}4^2$ va $1{,}5^2$ ni hisoblang. Kvadrati 2 ga teng son qaysi ikki son orasida?

b) $1{,}41^2$ va $1{,}42^2$ ni hisoblang.

c) Kalkulyatorda $\sqrt{2}$ ni toping. Raqamlarda takrorlanadigan guruh ko'rinyaptimi?

4-TOPSHIRIQ · Π VA $\frac{22}{7}$

Kalkulyatorda π tugmasini bosing, keyin $22 : 7$ ni hisoblang. Ikki son tengmi? Nechanchi raqamdan farq qiladi?

3-QISM

Tushunamiz

1-topshiriqdagi barcha sonlar — butun, kasr, aralash son, o'nli kasr — ikki butun sonning bo'linmasi ko'rinishida yozildi.

TA'RIF · RATSIONAL SONLAR

$\frac{a}{b}$ ko'rinishda yozish mumkin bo'lgan sonlar **ratsional sonlar** deb ataladi. Bu yerda a va b — butun sonlar va $b \neq 0$.

1-MISOL · BUTUN SONLAR

$6 = \frac{6}{1}$, $-8 = \frac{-8}{1}$, $0 = \frac{0}{3}$. Har qanday butun son — ratsional son.

2-MISOL · KASRLAR

$3\frac{1}{4} = \frac{13}{4}$, $0{,}35 = \frac{35}{100} = \frac{7}{20}$, $-1{,}2 = \frac{-6}{5}$. Chekli o'nli kasrlar oddiy kasr ko'rinishida yozilishi mumkin.

QOIDA · RATSIONAL SON O'NLI KASR KO'RINISHIDA

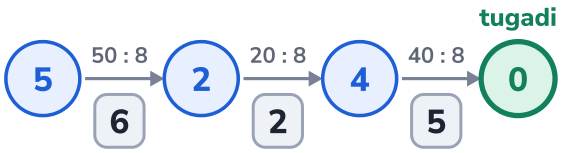
Har qanday $\frac{a}{b}$ kasr ko'rinishidagi ratsional son kasr suratini maxrajiga bo'lish natijasi sifatida o'nli kasr ko'rinishida yozilishi mumkin.

Chekli sondagi raqamlari bor o'nli kasr — **chekli o'nli kasr**. Raqamlari cheksiz ko'p bo'lib, davriy takrorlanib boradigan o'nli kasrlarni **davriy o'nli kasrlar** deb ataymiz. Takrorlanuvchi raqamlar qavs ichiga olinadi: $0{,}111\cdots = 0{,}(1)$.

Ratsional sonlar o'nli kasr ko'rinishida yozilganda ular chekli yoki davriy o'nli kasr bo'ladi.

3-MISOL · CHEKLI O'NLI KASR

Qoldiqlar: qoldiq × 10, keyin : 8



$\frac{5}{8} = 0{,}625$

Qoldiq 0 bo'ldi — raqamlar tugadi: chekli o'nli kasr.

4-MISOL · DAVR VERGULDAN KEYIN DARHOL

Qoldiqlar: qoldiq × 10, keyin : 11

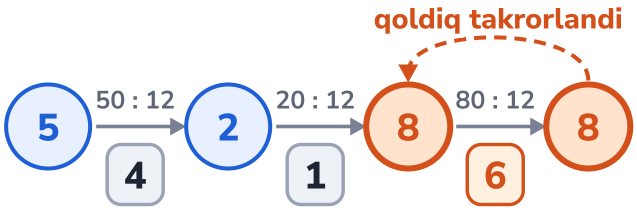


$\frac{7}{11} = 0{,}636363\ldots = 0{,}(63)$

Xuddi shunday: $\frac{25}{33} = 0{,}7575\cdots = 0{,}(75)$.

5-MISOL · DAVR KEYINROQ BOSHLANADI

Qoldiqlar: qoldiq × 10, keyin : 12



$\frac{5}{12} = 0{,}41666\ldots = 0{,}41(6)$

12 ga bo'lganda qoldiq 1 dan 11 gacha bo'ladi. Qoldiq 0 chiqmasa, ertami-kechmi bittasi qaytib keladi — raqamlar ham takrorlanadi. Qavsga faqat takrorlanadigan 6 olinadi.

TA'RIF · IRRATSIONAL VA HAQIQIY SONLAR

$\frac{a}{b}$ ko'rinishida yozish mumkin bo'lmagan sonlar ham mavjud (bu yerda a va b — butun sonlar, $b \neq 0$). Bu sonlarni chekli yoki davriy o'nli kasr ko'rinishida yozib bo'lmaydi. Bunday sonlar **irratsional sonlar** deyiladi. Masalan, π va $\sqrt{2}$ sonlari.

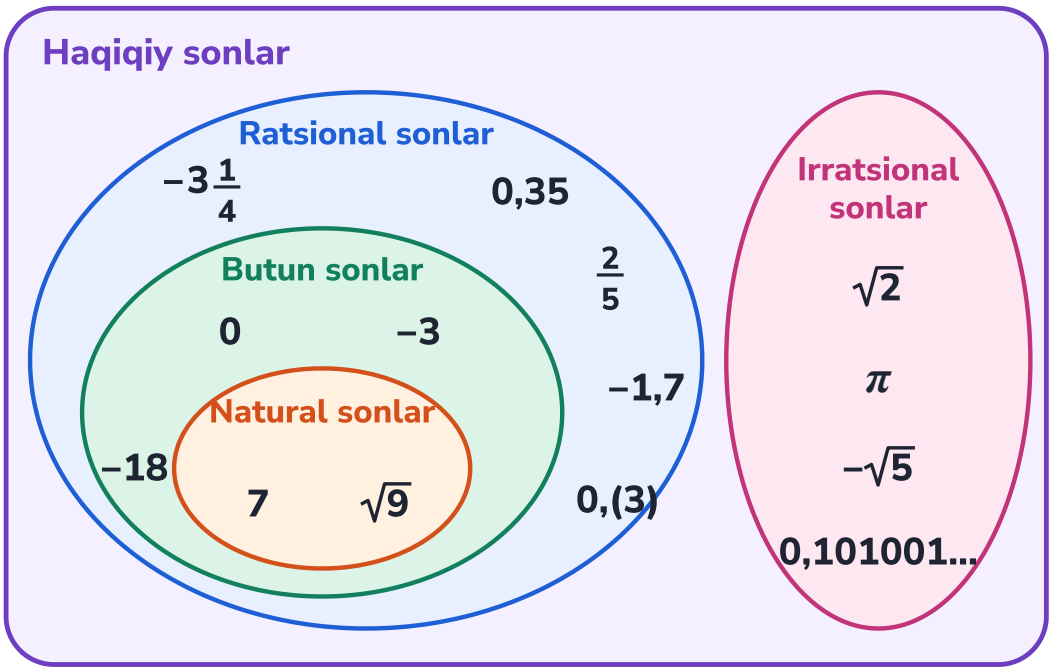
Ratsional va irratsional sonlar birgalikda **haqiqiy sonlarni** tashkil etadi. Har qanday haqiqiy son sonlar o'qida nuqta bilan belgilanadi.

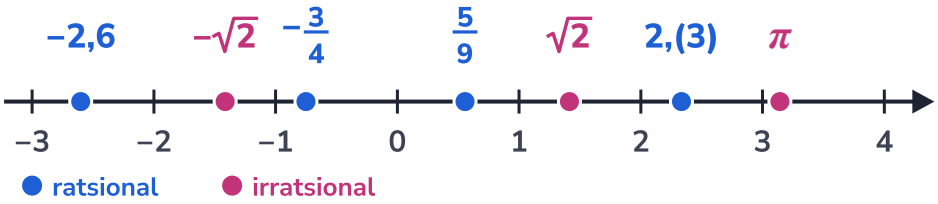
6-MISOL · IRRATSIONAL SON

$\pi = 3{,}14159\ldots$, $\sqrt{2} = 1{,}41421\ldots$, $\sqrt{5} = 2{,}23606\ldots$ — raqamlar cheksiz va takrorlanmaydi. To'la kvadrat bo'lmagan natural sonning kvadrat ildizi ($\sqrt{3}$, $\sqrt{7}$, ...) — irratsional.

7-MISOL · ILDIZ BOR, LEKIN RATSIONAL

$\sqrt{49} = 7$, $\sqrt[3]{-64} = -4$ (chunki $(-4)\cdot(-4)\cdot(-4) = -64$). Bular — butun sonlar, demak ratsional. Ildiz belgisi sonni irratsional qilmaydi.





1-QISMDAGI SAVOLLAR

1-savol. Yo'l $= \pi$ m. π — irratsional: uni oddiy kasr bilan aniq yozib bo'lmaydi. 3,14 va $\frac{22}{7} = 3{,}(142857)$ — π ning taqribiy qiymatlari.

2-savol. Yo'q. $\frac{2}{3} = 0{,}(6)$ — 6 raqami tugamaydi, ekrandagi oxirgi 7 — yaxlitlangan. Aniq javob: $\frac{2}{3}$ m.

4-QISM

Qo'llaymiz

1-MISOL · CHEKLI YOKI DAVRIY

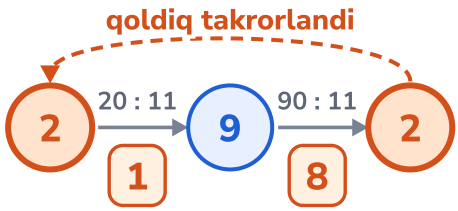
O'nli kasrga aylantiring: a) $\frac{7}{16}$ b) $-2\frac{1}{6}$ c) $\frac{13}{11}$

a) $7 : 16 = 0{,}4375$ — qoldiq 0 bo'ldi: chekli o'nli kasr.

b) Butun qism alohida, ishora oldinda qoladi: $\frac{1}{6} = 0{,}1666\ldots$, demak $-2\frac{1}{6} = -2{,}1(6)$ — davriy.

c) $13 : 11 = 1$, qoldiq 2. Keyin qoldiqlar 2, 9, 2, ... takrorlanadi: $\frac{13}{11} = 1{,}1818\cdots = 1{,}(18)$ — davriy.

Qoldiqlar: qoldiq × 10, keyin : 11



$\frac{13}{11}$ = 1,181818... = 1,(18)

2-MISOL · O'NLI KASRDAN QISQARMAS KASRGA

a) 0,45 b) −6,75 c) 0,888

a) Ikki xona — maxraj 100: $0{,}45 = \frac{45}{100} = \frac{9}{20}$ (5 ga qisqartirdik).

b) Butun qism alohida: $-6{,}75 = -6\frac{75}{100} = -6\frac{3}{4}$.

c) 0,888 — chekli kasr (uch xona): $\frac{888}{1000} = \frac{111}{125}$. Bu $0{,}(8)$ emas!

3-MISOL · RATSIONALMI, IRRATSIONALMI?

a) $\sqrt[3]{-125}$ b) $-4{,}1(7)$ c) $\pi + 2$ d) $0{,}(6) \cdot 3$ e) $\sqrt{20}$

a) $(-5) \cdot (-5) \cdot (-5) = -125$, demak $\sqrt[3]{-125} = -5$ — butun son, **ratsional**.

b) $-4{,}1777\ldots$ — davriy o'nli kasr, **ratsional**.

c) $\pi + 2 = 5{,}14159\ldots$ — π ning raqamlari takrorlanmaydi, 2 ni qo'shsak ham shunday: **irratsional**.

d) $0{,}(6) = \frac{2}{3}$ (1-qismdagi 2-savol), $\frac{2}{3} \cdot 3 = 2$ — butun son, **ratsional**.

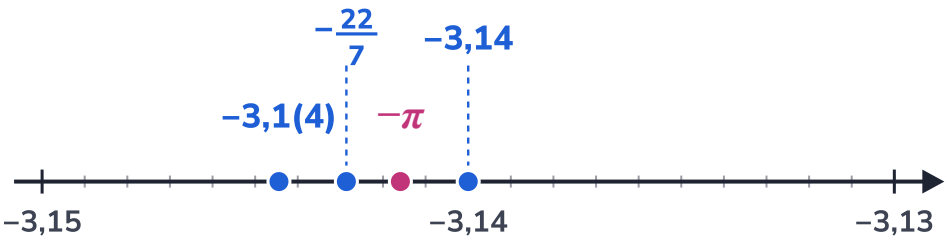
e) $16 < 20 < 25$: 20 to'la kvadrat emas, $\sqrt{20} = 4{,}472\ldots$ — **irratsional**.

4-MISOL · TARTIBLASH

O'sib borish tartibida yozing: $-\pi$; $-3{,}1(4)$; $-\frac{22}{7}$; $-3{,}14$.

1. Hammasini bir xil xonagacha yozamiz: $-\pi = -3{,}14159\ldots$, $-3{,}1(4) = -3{,}14444\ldots$, $-\frac{22}{7} = -3{,}14285\ldots$, $-3{,}14 = -3{,}14000$.

2. Manfiy sonlarda moduli katta son kichik bo'ladi. Modullar: $3{,}14444 > 3{,}14285 > 3{,}14159 > 3{,}14$.



3. $-3{,}1(4) < -\frac{22}{7} < -\pi < -3{,}14$.

5-MISOL · DAVRDA NECHTA RAQAM BO'LISHI MUMKIN?

Butun sonni 7 ga bo'lganda davrda eng ko'pi bilan nechta raqam bo'ladi?

Qoldiqlar: qoldiq × 10, keyin : 7



$\frac{1}{7}$ = 0,1428571428571... = 0,(142857)

1. 7 ga bo'lganda qoldiq 0, 1, 2, 3, 4, 5, 6 bo'lishi mumkin. 0 chiqsa, bo'lish tugaydi.
2. Qolgan 6 xil qoldiq bor. Ko'pi bilan 6 qadamdan keyin qoldiq takrorlanadi: davrda ko'pi bilan 6 ta raqam.
3. $\frac{1}{7}$ da hamma 6 qoldiq chiqdi: $\frac{1}{7} = 0,(142857)$. Javob: 6.

6-MISOL · DAVRIY KASRDAN ODDIY KASRGA

$\frac{1}{9} = 0,(1)$ va $\frac{1}{90} = 0,0(1)$ dan foydalanib, a) $0,(4)$ b) $0,5(4)$ ni oddiy kasr ko'rinishida yozing.

a) $0,444\ldots - 0,111\ldots$ ning 4 baravari: $0,(4) = 4 \cdot \frac{1}{9} = \frac{4}{9}$.

b) $0,5(4) - 0,5 = 0,0444\cdots = 4 \cdot \frac{1}{90} = \frac{4}{90}$. Demak $0,5(4) = \frac{5}{10} + \frac{4}{90} = \frac{45}{90} + \frac{4}{90} = \frac{49}{90}$.

✓ $49 : 90 = 0,5444\ldots$

7-MISOL · IKKI SON ORASIDAGI SON

$\frac{2}{3}$ va $\frac{3}{4}$ orasida: a) chekli o'nli kasr; b) davriy o'nli kasr; c) irratsional son toping.

1. O'nli kasrga aylantiramiz: $\frac{2}{3} = 0{,}666\ldots$, $\frac{3}{4} = 0{,}75$.
2. a) $0{,}7$ b) $0{,}7(2) = 0{,}7222\ldots$ c) $0{,}71011011101111\ldots$ — har safar birlar bittaga ko'payadi, takrorlanadigan guruh yo'q.

Boshqa javoblar ham to'g'ri: masalan, 0,68 yoki 0,(70).

8-MISOL · ROSTMI, YOLG'ONMI?

«Ikki irratsional sonning ayirmasi har doim irratsional son.»

1. Tekshiramiz: $\pi + 1 = 4{,}14159\ldots$ va π — ikkalasi irratsional (3-misol, c). Ayirmasi $(\pi + 1) - \pi = 1$ — ratsional.
2. Bitta zid misol topildi — tasdiq **yolg'on**. Bunday misol **kontrmisol** deyiladi.



Kalkulyator ekranidagi oxirgi raqam yaxlitlangan bo'ladi. Davr bor-yo'qligini qoldiqlarga qarab aniqlang.

DIQQAT!

1. Raqamlari tugagan kasr davriy emas: ✗ 0,52727 — davriy kasr ✓ **0,52727 — chekli; davriysi** $0{,}5(27) = 0{,}52727\ldots$
2. Davr qavsi faqat takrorlanadigan raqamlarda: ✗ $\frac{5}{22} = 0{,}(227)$ ✓ $\frac{5}{22} = 0{,}22727\cdots = 0{,}2(27)$
3. ✗ $\pi = \frac{22}{7}$, demak π ratsional ✓ $\frac{22}{7} \approx \pi$; π **— irratsional**
4. ✗ $\sqrt{36}$ — irratsional ✓ $\sqrt{36} = 6$ **— ratsional**
5. ✗ $2 : 3 = 0{,}6666666667$ ✓ $2 : 3 = 0{,}(6)$, **ekrandagi 7 — yaxlitlangan**

Mashqlar

1. Sonlarni $\frac{a}{b}$ ko'rinishida yozing (a, b — butun sonlar, $b \neq 0$).
   a) −7 b) 0 c) 15 d) $4\frac{2}{9}$ e) 0,6 f) −2,05

2. O'nli kasrga aylantiring. Chekli yoki davriy o'nli kasr ekanini aniqlang.
   a) $\frac{9}{20}$ b) $-3\frac{3}{8}$ c) $\frac{5}{9}$
   d) $-\frac{14}{11}$ e) $-\frac{11}{12}$ f) $\frac{7}{40}$

3. Qisqarmas oddiy kasr yoki aralash son ko'rinishida yozing.
   a) 0,48 b) 0,888 c) 4,35 d) −6,24

4. Qisqarmas oddiy kasr, aralash son yoki butun son ko'rinishida yozing.
   a) $\frac{24}{36}$ b) $\frac{-9}{0{,}75}$ c) $5\frac{21}{84}$ d) −12,5

5. Kalkulyatordan foydalanib, chekli yoki davriy o'nli kasr ko'rinishida yozing.
   a) $\left(2\frac{1}{2}\right)^2$ b) $\left(-\frac{1}{3}\right)^3$ c) $-\left(\frac{5}{3}\right)^2$ d) $\left(1\frac{1}{5}\right)^2$

6. a) Qavs yordamida qisqa yozing: 0,272727...; 3,15555...; 0,408408408...; 2,0191919...
   b) Verguldan keyin 7 ta raqamini yozing: $0{,}(36)$; $1{,}2(05)$.

7. Ratsional yoki irratsional son ekanini aniqlang.
   a) $-\frac{5}{4}$ b) $\sqrt{64}$ c) $\sqrt[3]{-27}$ d) $\pi - 2$
   e) $0{,}(3) \cdot 6$ f) $\sqrt{11}$ g) $2{,}7(3)$ h) $3{,}1416$

8. Har bir son qaysi to'plamlarga kiradi: natural, butun, ratsional, irratsional, haqiqiy?
   12; −6; 0; $\frac{3}{5}$; $-1{,}(4)$; $\sqrt{5}$; $\sqrt{25}$

9. O'sib borish tartibida yozing: $-0{,}(61)$; $-0{,}6(1)$; $-0{,}61(6)$; $-\frac{8}{13}$; −0,61.

10. a) Kamayib borish tartibida yozing: $\frac{5}{9}$; $0{,}5(4)$; 0,55; $\frac{6}{11}$.
    b) O'sib borish tartibida yozing: $-\frac{2}{3}$; $-0{,}(6)$; −0,67; $-0{,}6(7)$.

11. Sonlarni sonlar o'qida belgilang va o'sib borish tartibida yozing: −3; $-\sqrt{7}$; $2{,}(6)$; $\frac{8}{3}$; π; 2,6.

12. Butun sonni a) 11 ga; b) 13 ga bo'lganda hosil bo'lgan o'nli kasrning davrida eng ko'pi bilan nechta raqam bo'ladi? Javobingizni tushuntiring.

13. Rostmi, yolg'onmi? Yolg'on bo'lsa, kontrmisol keltiring.
    a) Har qanday butun son — ratsional son.
    b) Har qanday ratsional son — butun son.
    c) Har qanday chekli o'nli kasr — ratsional son.
    d) Ikki davriy o'nli kasrning yig'indisi har doim davriy o'nli kasr bo'ladi.
    e) Kvadrati butun son bo'lgan har qanday son butun son bo'ladi.
    f) Ikki butun sonning bo'linmasi (bo'luvchi 0 emas) — ratsional son.

14. $\frac{1}{4} < a < \frac{1}{3}$ bo'ladigan a sonni toping: a) a — chekli o'nli kasr; b) a — davriy o'nli kasr; c) a — irratsional son.

15. a) Verguldan keyin faqat 4 va 9 raqamlari keladigan irratsional son yozing.
    b) Aziz: «$0,(49)$ ham faqat 4 va 9 dan tuzilgan, demak u ham irratsional». Aziz haqmi?

16. a) $\frac{1}{90}$ ni davriy o'nli kasr ko'rinishida yozing.
    b) $0,6(2) - 0,6$ ni hisoblang. Javobni davriy o'nli kasr ko'rinishida yozing.
    c) a) banddagi javobdan foydalanib, $0,6(2)$ ni qisqarmas oddiy kasr ko'rinishida yozing.

17. Manfiy ratsional sonni $\frac{2}{9}$, $\frac{4}{15}$ va $\frac{8}{21}$ ning har biriga bo'lganda butun son chiqdi. Bu sonning eng katta qiymatini toping.
    Yordam: son har bir kasrning butun songa ko'paytmasi bo'lishi kerak.

5-QISM

Tekshiramiz

1. $\frac{a}{b}$ ko'rinishida yozing: −11; $3\frac{4}{7}$; 0,9; −1,6.

2. O'nli kasrga aylantiring, chekli yoki davriy ekanini ayting: a) $\frac{11}{25}$ b) $\frac{4}{9}$ c) $-\frac{7}{6}$ d) $\frac{2}{37}$

3. Qisqarmas oddiy kasr yoki aralash son ko'rinishida yozing: a) 0,85 b) −2,125 c) 0,555

4. Qavs yordamida yozing: a) 0,818181... b) 5,2333... c) 0,123123...
5. Ratsional sonlar va irratsional sonlar guruhiga ajrating: $\sqrt{100}$; $\sqrt[3]{-8}$; $\pi + 5$; $1,(6)$; $\sqrt{13}$; $-\frac{7}{9}$.
6. O'sib borish tartibida yozing: $0,(36)$; $0,3(6)$; 0,36; $\frac{4}{11}$; $0,36(3)$.
7. Kalkulyatordan foydalanib, kamayib borish tartibida yozing: $\sqrt{10}$; $3,(16)$; 3,16; $3,1(6)$.
8. 0,1 va 0,2 orasidagi a) chekli o'nli kasrni; b) davriy o'nli kasrni; c) irratsional sonni yozing.
9. Rostmi, yolg'onmi? a) Har qanday natural son — butun son. b) Har qanday irratsional son — haqiqiy son. c) Har qanday haqiqiy son — ratsional son. d) $\pi \cdot 0$ — irratsional son.
10. $\frac{1}{9} = 0,(1)$ va $\frac{1}{90} = 0,0(1)$ dan foydalanib, oddiy kasr ko'rinishida yozing: a) $0,(7)$ b) $0,2(7)$
11. Qaysi biri davriy o'nli kasr emas: $0,(25)$; 0,25252; $0,2(5)$; $\frac{25}{99}$?
12. **Xatoni toping.**
    a) Jasur: «$\frac{22}{7} = \pi$, shuning uchun π — ratsional son.»
    b) Nodira: «$\sqrt{49}$ — irratsional son, chunki ildiz belgisi bor.»
    c) Behruz: «$\frac{5}{22} = 0,(227)$.»

- [ ] Men sonni $\frac{a}{b}$ ko'rinishida yozib, ratsional ekanini ko'rsata olaman.
- [ ] Men oddiy kasrni o'nli kasrga aylantirib, chekli yoki davriy ekanini aniqlay olaman va davrni qavsga olib yoza olaman.
- [ ] Men ratsional va irratsional sonlarni ajrata olaman (π, $\sqrt{2}$ — irratsional, $\sqrt{49}$ — ratsional).
- [ ] Men haqiqiy sonlarni sonlar o'qida belgilab, tartiblay olaman.

Javoblar

4-qism · Mashqlar

1-masala

a) $\frac{-7}{1}$ b) $\frac{0}{1}$ c) $\frac{15}{1}$ d) $\frac{38}{9}$ e) $\frac{3}{5}$ f) $-2{,}05 = -2\frac{1}{20} = \frac{-41}{20}$.

2-masala

a) 0,45 — chekli. b) −3,375 — chekli. c) $0{,}(5)$ — davriy. d) $-1{,}(27)$ — davriy. e) $-0{,}91(6)$ — davriy. f) 0,175 — chekli.

3-masala

a) $\frac{48}{100} = \frac{12}{25}$. b) $\frac{888}{1000} = \frac{111}{125}$. c) $4\frac{35}{100} = 4\frac{7}{20}$. d) $-6\frac{24}{100} = -6\frac{6}{25}$.

4-masala

a) $\frac{2}{3}$ (12 ga qisqartirdik). b) $-9 : 0{,}75 = -900 : 75 = -12$. c) $\frac{21}{84} = \frac{1}{4}$, $5\frac{1}{4}$. d) $-12\frac{1}{2}$.

5-masala

a) $\left(\frac{5}{2}\right)^2 = \frac{25}{4} = 6{,}25$ — chekli. b) $-\frac{1}{27} = -0{,}(037)$ — davriy. c) $-\frac{25}{9} = -2{,}(7)$ — davriy. d) $\left(\frac{6}{5}\right)^2 = \frac{36}{25} = 1{,}44$ — chekli.

6-masala

a) $0{,}(27)$; $3{,}1(5)$; $0{,}(408)$; $2{,}0(19)$. b) 0,3636363; 1,2050505.

7-masala

a) ratsional. b) $\sqrt{64} = 8$ — ratsional. c) $\sqrt[3]{-27} = -3$ — ratsional. d) irratsional. e) $0{,}(3) = \frac{1}{3}$, $\frac{1}{3} \cdot 6 = 2$ — ratsional. f) 11 to'la kvadrat emas — irratsional. g) davriy — ratsional. h) chekli o'nli kasr — ratsional (bu π emas, π ning taqribiy qiymati).

8-masala

12 — natural, butun, ratsional, haqiqiy. −6 va 0 — butun, ratsional, haqiqiy. $\frac{3}{5}$ va $-1,(4)$ — ratsional, haqiqiy. $\sqrt{5}$ — irratsional, haqiqiy. $\sqrt{25}=5$ — natural, butun, ratsional, haqiqiy.

9-masala

Modullarini bir xil xonagacha yozamiz: $0,(61)=0,61616\ldots$, $0,6(1)=0,61111\ldots$, $0,61(6)=0,61666\ldots$, $\frac{8}{13}=0,61538\ldots$, $0,61000$. Manfiy sonlarda moduli katta son kichik:

$$-0,61(6)<-0,(61)<-\frac{8}{13}<-0,6(1)<-0,61.$$

10-masala

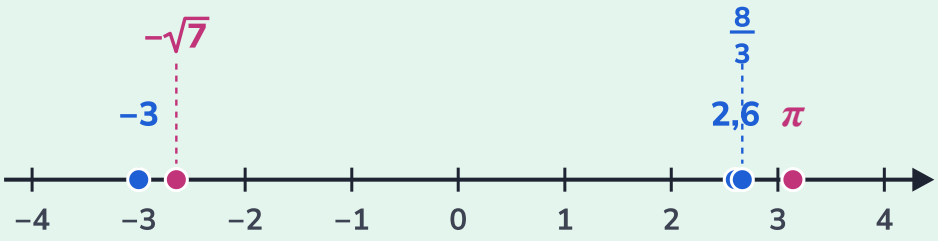
a) $\frac{5}{9}=0,5555\ldots$, $0,5(4)=0,5444\ldots$, $0,55=0,5500$, $\frac{6}{11}=0,5454\ldots$: $\frac{5}{9}>0,55>\frac{6}{11}>0,5(4)$.

b) $\frac{2}{3}=0,(6)=0,6666\ldots$, $0,67=0,6700$, $0,6(7)=0,6777\ldots$: $-0,6(7)<-0,67<-\frac{2}{3}=-0,(6)$.

11-masala

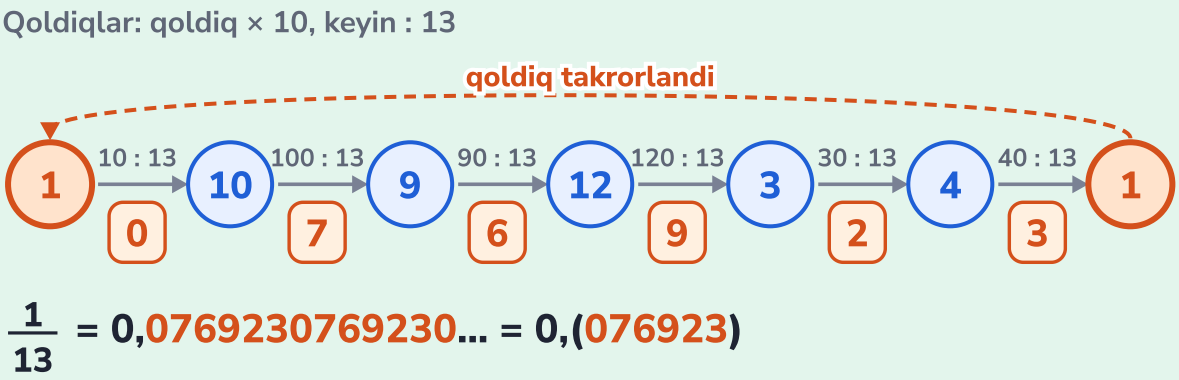
$-\sqrt{7}\approx-2,65$, $2,(6)=\frac{8}{3}=2,666\ldots$, $\pi\approx3,14$.

$-3<-\sqrt{7}<2,6<2,(6)=\frac{8}{3}<\pi$.



12-masala

a) 11 ga bo'lganda qoldiq 1 dan 10 gacha bo'lishi mumkin, lekin zanjir har doim qisqa: 1 dan keyin 10, keyin yana 1; 3 dan keyin 8, keyin yana 3. Davrda ko'pi bilan 2 ta raqam: $\frac{1}{11} = 0,(09)$, $\frac{3}{11} = 0,(27)$. b) Qoldiq 1 dan 12 gacha bo'lishi mumkin, lekin har safar 6 qadamda takrorlanadi: $\frac{1}{13} = 0,(076923)$, $\frac{2}{13} = 0,(153846)$. Eng ko'pi bilan 6 ta raqam.



$\frac{1}{13}$ = 0,0769230769230... = 0,(076923)

13-masala

a) Rost: $n = \frac{n}{1}$. b) Yolg'on: $\frac{1}{2}$ — ratsional, lekin butun emas. c) Rost: masalan, $0,37 = \frac{37}{100}$. d) Yolg'on: $0,(3) + 0,(6) = \frac{1}{3} + \frac{2}{3} = 1$ — butun son. e) Yolg'on: $\sqrt{2} \cdot \sqrt{2} = 2$, lekin $\sqrt{2}$ butun emas. f) Rost: bu $\frac{a}{b}$ ko'rinishidagi son — ratsional sonning ta'rifi.

14-masala

$\frac{1}{4} = 0,25$, $\frac{1}{3} = 0,(3)$. Masalan: a) 0,3; b) $0,(27) = 0,2727\ldots$; c) $0,2101001000100001\ldots$ yoki $\frac{\pi}{10} = 0,314159\ldots$

15-masala

a) Masalan: $0,494994999499994\ldots$ — har safar 9 lar bittaga ko'payadi, takrorlanadigan guruh yo'q. b) Yo'q: $0,(49) = 0,4949\ldots$ — davriy o'nli kasr, demak ratsional ($\frac{49}{99}$). Irratsional bo'lishi uchun raqamlar takrorlanmasligi kerak.

16-masala

a) $1:90=0{,}0111\cdots=0{,}0(1)$. b) $0{,}6222\cdots-0{,}6=0{,}0(2)$. c) $0{,}0(2)=2\cdot\dfrac{1}{90}=\dfrac{2}{90}$, $0{,}6(2)=\dfrac{6}{10}+\dfrac{2}{90}=\dfrac{54}{90}+\dfrac{2}{90}=\dfrac{56}{90}=\dfrac{28}{45}$.

17-masala

Son $\dfrac{2}{9}$ ga bo'linsa, uni $\dfrac{2}{9}\cdot k$ deb yozish mumkin. Surat 2, 4, 8 ning har biriga bo'linishi kerak — kamida 8; maxraj 9, 15, 21 ning har birida bor bo'lishi kerak — ko'pi bilan 3 (umumiy bo'luvchi). Eng kichik musbati $\dfrac{8}{3}$, eng katta manfiysi $-\dfrac{8}{3}$.
Tekshiramiz: $-\dfrac{8}{3}:\dfrac{2}{9}=-12$, $-\dfrac{8}{3}:\dfrac{4}{15}=-10$, $-\dfrac{8}{3}:\dfrac{8}{21}=-7$.

5-qism · Tekshiramiz

1-masala

$\dfrac{-11}{1};\ \dfrac{25}{7};\ \dfrac{9}{10};\ \dfrac{-8}{5}$.

2-masala

a) $0{,}44$ — chekli. b) $0{,}(4)$ — davriy. c) $-1{,}1(6)$ — davriy. d) $0{,}(054)$ — davriy.

3-masala

a) $\dfrac{17}{20}$. b) $-2\dfrac{1}{8}$. c) $\dfrac{555}{1000}=\dfrac{111}{200}$.

4-masala

a) $0{,}(81)$. b) $5{,}2(3)$. c) $0{,}(123)$.

5-masala

Ratsional: $\sqrt{100}=10$, $\sqrt[3]{-8}=-2$, $1{,}(6)$, $-\dfrac{7}{9}$. Irratsional: $\pi+5$, $\sqrt{13}$.

6-masala

$0{,}36<0{,}36(3)<0{,}(36)=\dfrac{4}{11}<0{,}3(6)$ ($0{,}36333\cdots<0{,}36363\cdots<0{,}36666\ldots$).

7-masala

$3{,}1(6) = 3{,}1666\ldots$, $\sqrt{10} = 3{,}1622\ldots$, $3{,}(16) = 3{,}1616\ldots$, $3{,}16 = 3{,}1600$: $3{,}1(6) > \sqrt{10} > 3{,}(16) > 3{,}16$.

8-masala

Masalan: a) 0,15; b) $0{,}1(5)$; c) $0{,}1010010001\ldots$

9-masala

a) Rost. b) Rost. c) Yolg'on: $\sqrt{2}$ — haqiqiy, lekin ratsional emas. d) Yolg'on: $\pi \cdot 0 = 0$.

10-masala

a) $0{,}(7) = 7 \cdot \dfrac{1}{9} = \dfrac{7}{9}$. b) $0{,}2(7) = \dfrac{2}{10} + \dfrac{7}{90} = \dfrac{18}{90} + \dfrac{7}{90} = \dfrac{25}{90} = \dfrac{5}{18}$.

11-masala

0,25252 — raqamlari tugagan, chekli kasr. $\dfrac{25}{99} = 0{,}(25)$ — davriy.

12-masala

a) $\dfrac{22}{7} = 3{,}(142857)$ — π ning taqribiy qiymati, π esa irratsional. b) $\sqrt{49} = 7$ — ratsional. c) $\dfrac{5}{22} = 0{,}22727\cdots = 0{,}2(27)$: birinchi 2 takrorlanmaydi.